

FCU3

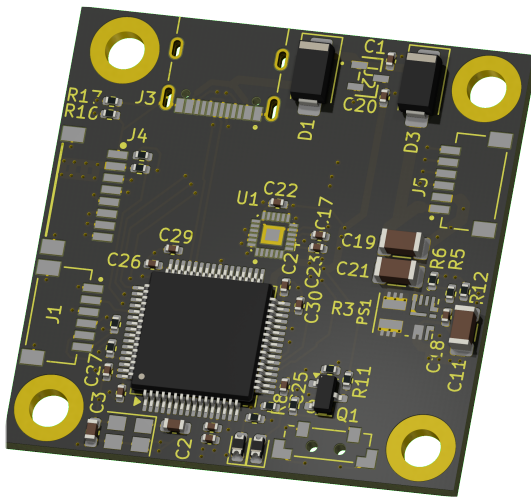
Variant: PRELIMINARY

2025-12-06

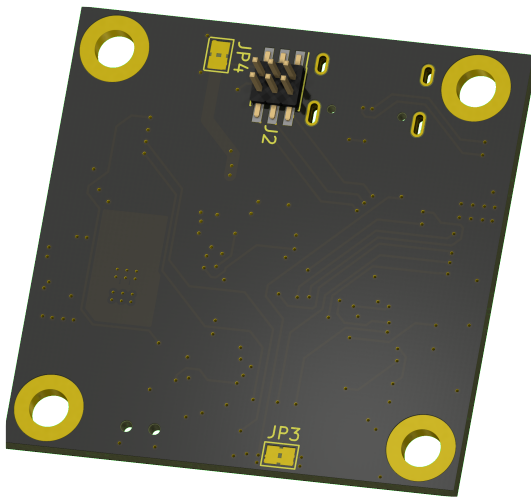
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 06-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

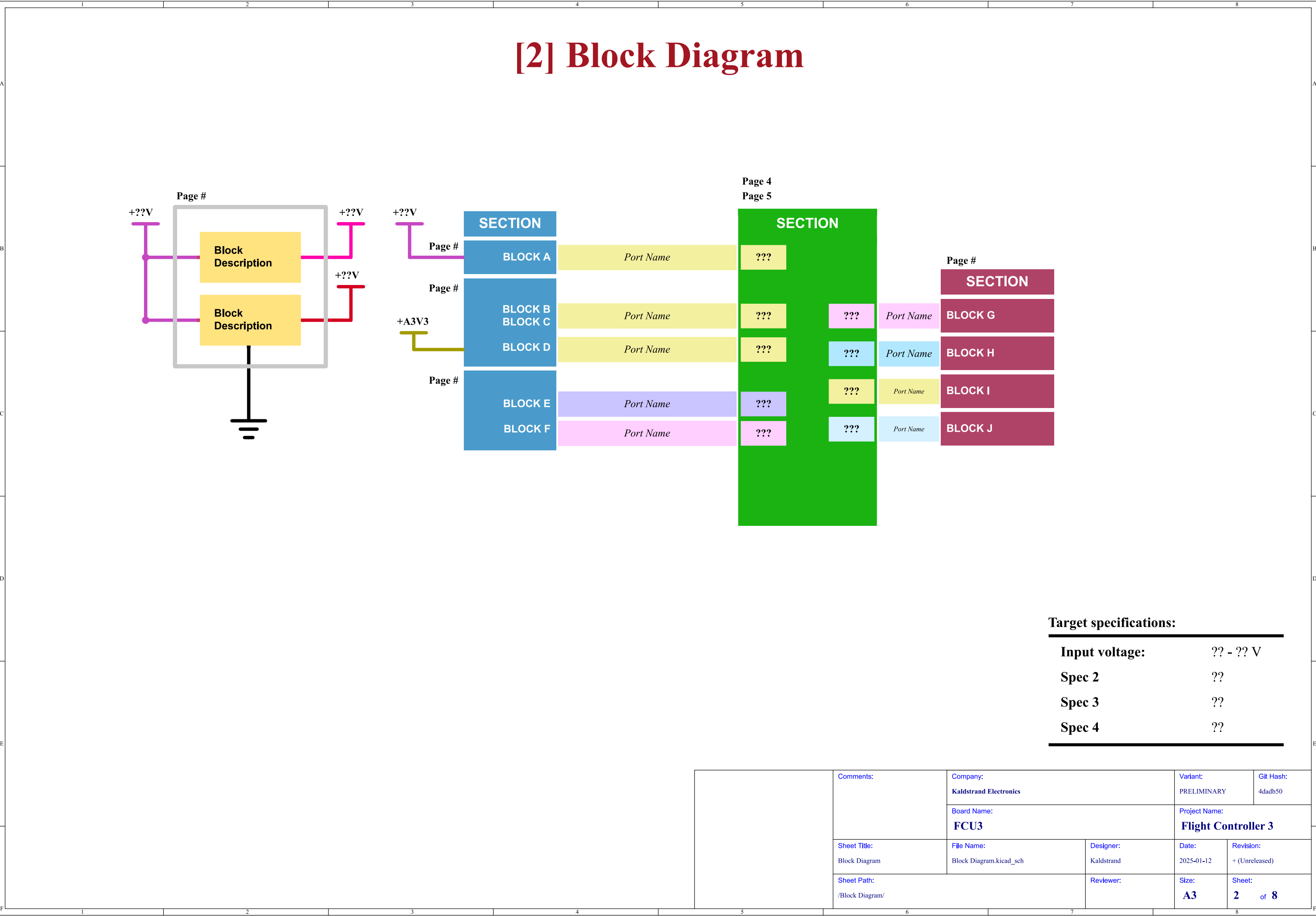
DESIGN NOTE:

Example text for critical design notes.

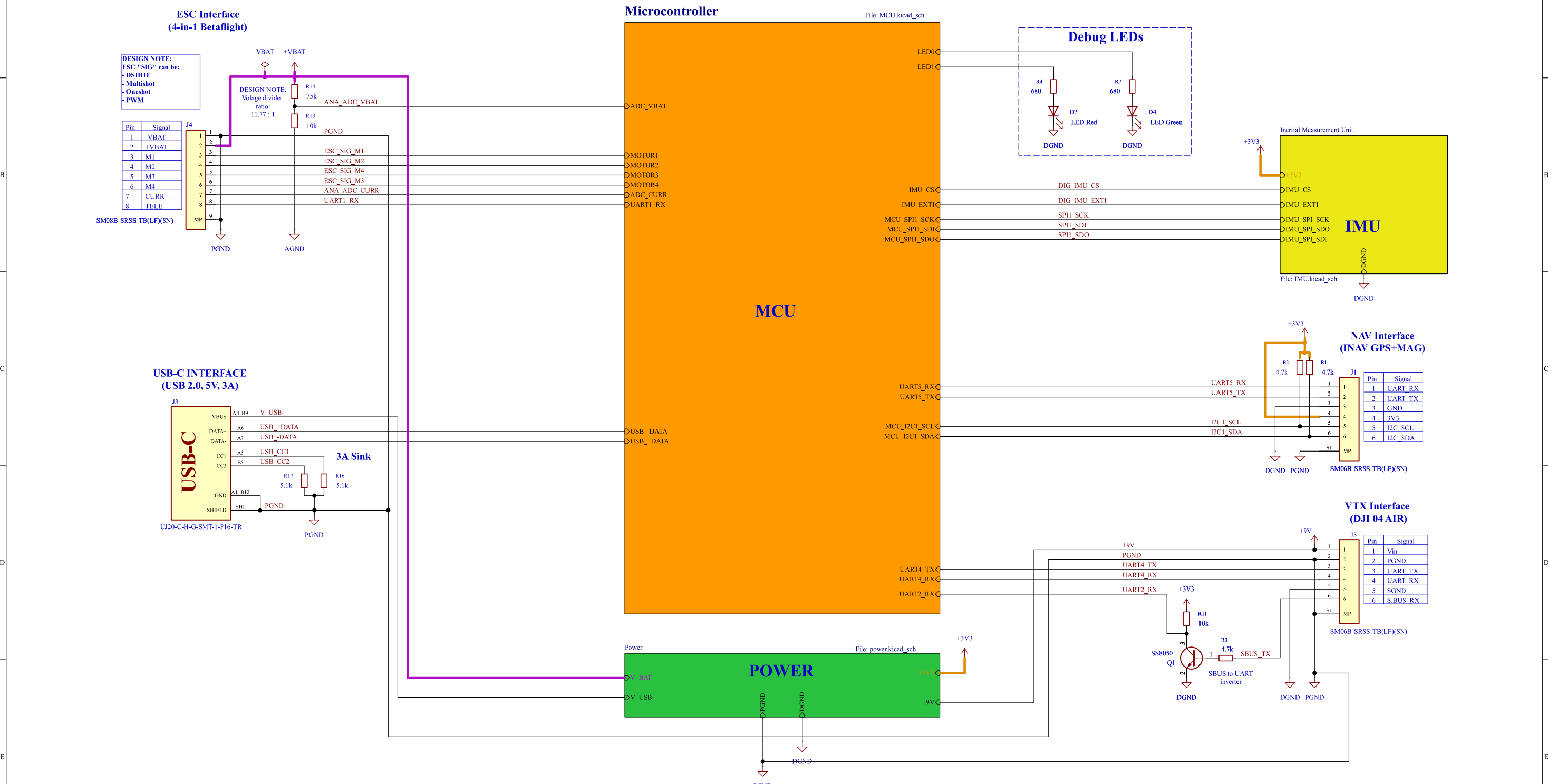
LAYOUT NOTE:

Example text for critical layout guidelines.

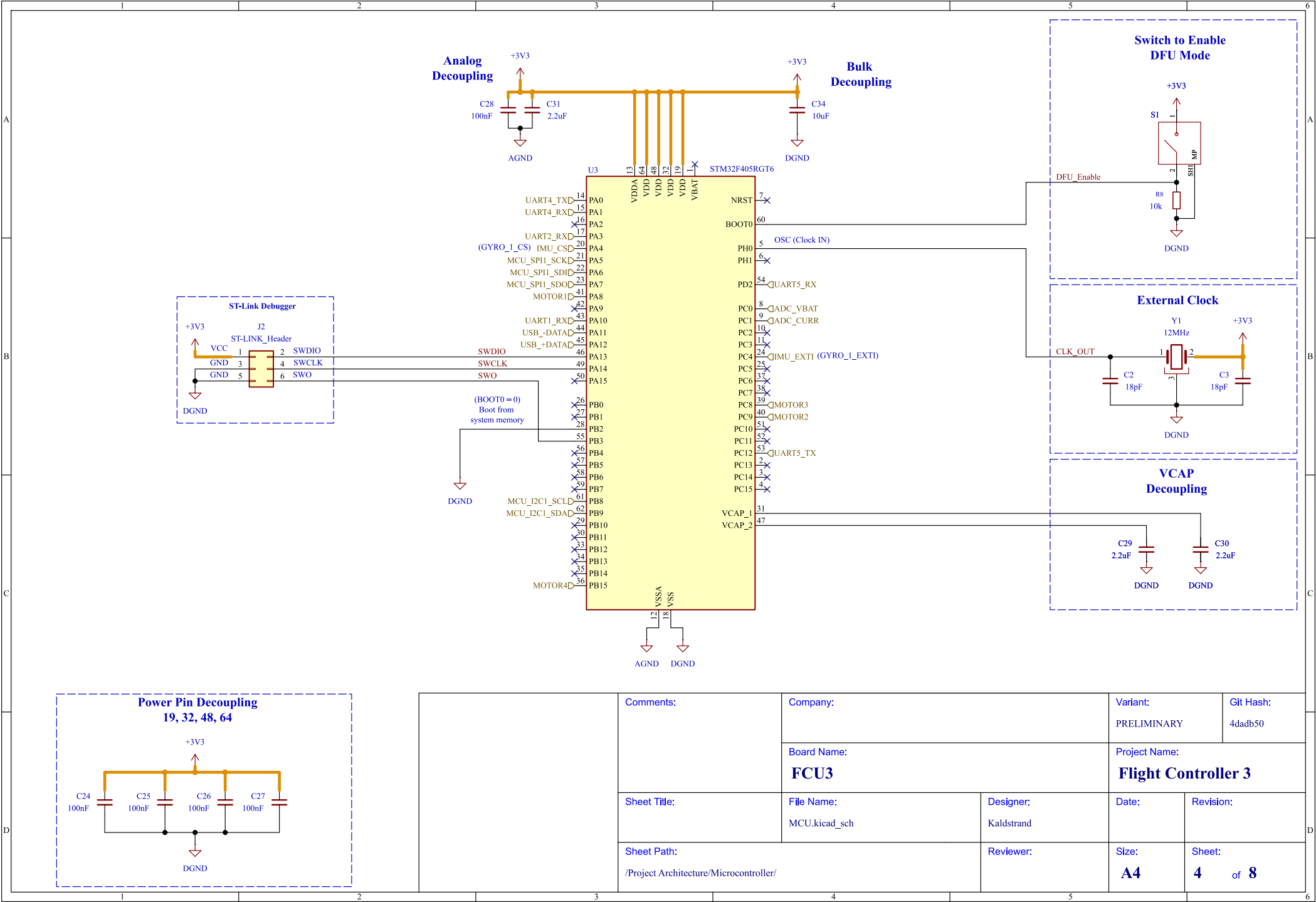
	Comments:	Company: Kaldstrand Electronics		Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: FCU3.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 8

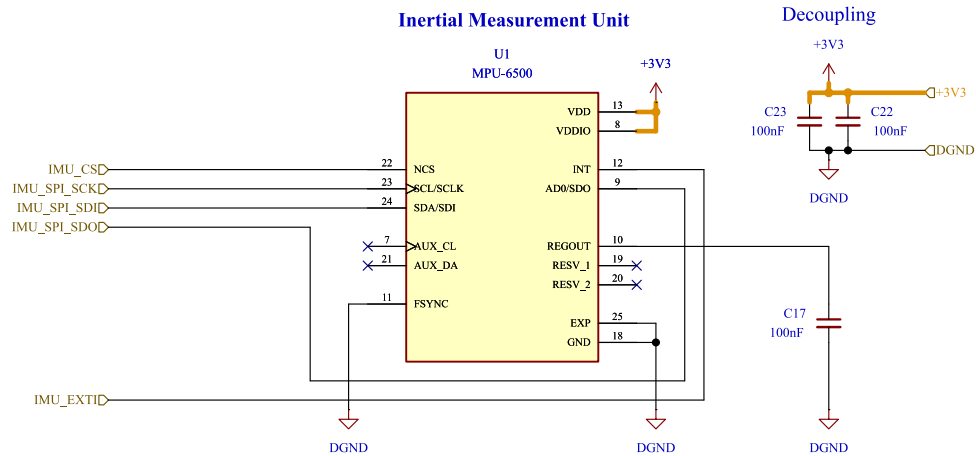


[3] Project Architecture



Comments:		Company:		Variant:	Git Hash:
		Kaldstrand Electronics		PRELIMINARY	4dad50
Sheet Title:		Board Name:		Project Name:	
		FCU3		Flight Controller 3	
Sheet Path:		File Name:		Date:	Revision:
		Project Architecture.kicad_sch		2025-01-12	+(Unreleased)
		Designer:		Size:	Sheet:
		Kaldstrand		A3	3 of 8
		Reviewer:			





	Comments:	Company:	Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4 Sheet: 5 of 8

To-Do:

- Add all settings used in Vbat to 9V converter.
- Add all settings used in 3.3V LDO.

10-26V to 9V2A Step-Down Converter
File: stepdown_10-26vin_9v2a.kicad_sch

Vin

+9V

GND

V_USB

5V @ (0, 500, 3000)mA

10-25.2V @ (0, 3, 100)A

D3
SS14-M3/SAT

D1
SS14-M3/SAT

C1
1uF

LDO
3V3 @ (0, 50, 200)mA
U2
AP7375-33SA-7

VIN

VOUT

GND

C20
1uF

+3V3

DGND

JP4

DESIGN NOTE:
JP4 is the Power Ground (PGND) to
Digital Ground (DGND) connection.

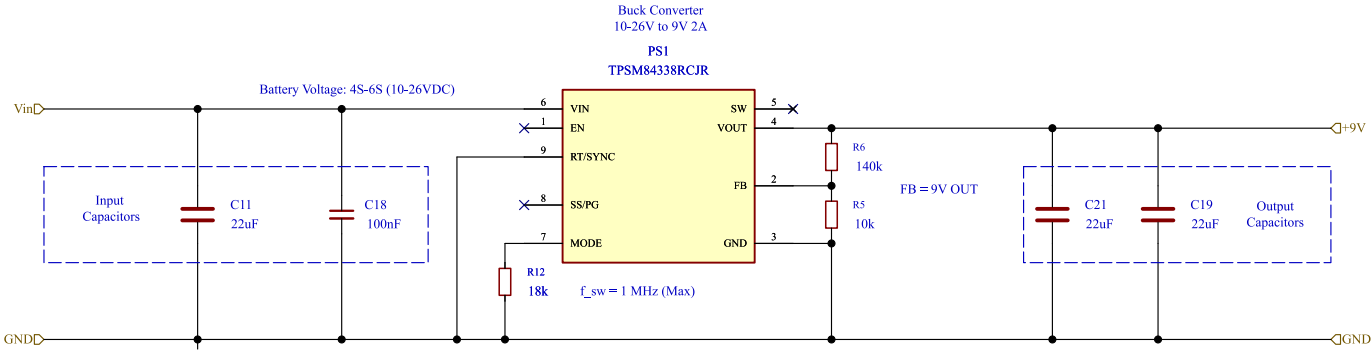
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				PRELIMINARY	4dadb50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name: power.kicad_sch	Designer: Kaldstrand	Date:	Revision:
	Sheet Path: /Project Architecture/Power/	Reviewer:	Size: A4	Sheet: 6 of 8	

	Comments:	Company:		Variant:	Git Hash:
				PRELIMINARY	4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
		power.kicad_sch	Kaldstrand		
Sheet Path:			Reviewer:	Size:	Sheet:
/Project Architecture/Power/				A4	6 of 8

DESIGN NOTE:
 $R_{FBT} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBT} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBT} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:	Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: stepdown_10-26vin_9v2a.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Power/10-26V to 9V2A Step-Down Converter/	Reviewer:	Size: A4	Sheet: 8 of 8

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8